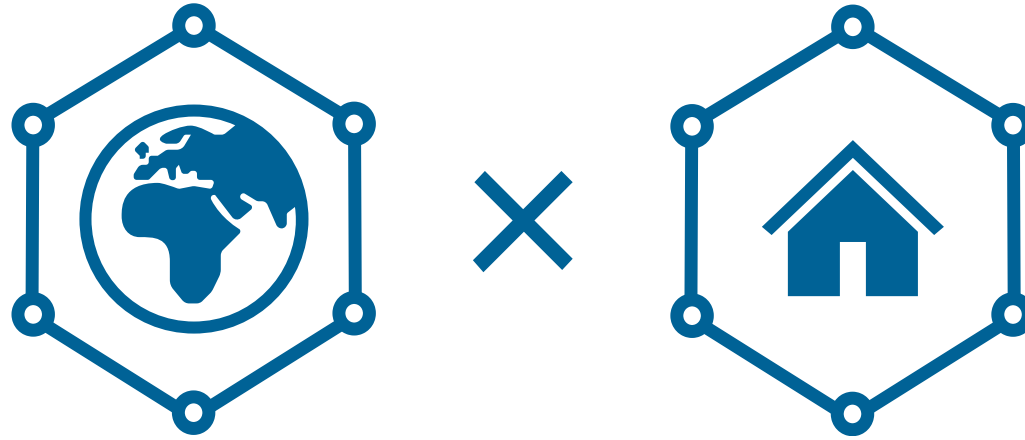




Applications and Architectures
Building KG-based
Enterprise AI Architectures

Emanuel Sallinger



AI-based Applications

- great hype
- proliferation of AI applications
- companies wish to catch this hype and **innovate processes** and build new **IT systems**
- avoid IT assets from becoming a **potpourri of heterogeneous applications**
 - applications do **not cooperate**
 - **duplicated** solutions
 - degenerate into **disconnected silos**



AI Strategy

- what are our top **business priorities**?
- what **problems** do we want to solve?
- how can AI help achieve **strategic business goals**?

- **Strategic AI priorities**

- improvement of products
- improving business processes and functions
- reducing human burden





AI Strategy

- **Short-term AI adoption priorities**
 - catch the low-hanging fruits
 - for example:
 - small projects, which could benefit from AI
 - small business processes, with some AI impact
- **Data strategy**
 - data-driven AI, implies the need for a data strategy
 - do we have the right data to implement our AI strategy?
 - access to them?
 - rights to read/use?
 - do we have sufficient business involvement?
 - do we have the right volume?
 - do we have the right quality?





KG-based IT Architecture

- The IT architecture of AI-driven knowledge companies should be founded on a **Knowledge Graph (KG)** that **models the enterprise context**.
- Enterprise KGs must be deployed into a **Knowledge Graph Management System (KGMS)**
 - provides a **language and a formalism** for representation and reasoning on KGs
 - can access a **rich set of data sources** (including big data)
 - can embed procedural **third-party code**
 - provides reasoning sources via a **rich set of API**
- **Access to Big Data**
 - streamable approach
 - lazy mode
- **Third-party code**
 - machine learning
 - text mining
 - NLP
 - data analytics & visualisation





Towards Graphs for Enterprise AI

A Knowledge Graph is a **semi-structured** data model characterised by three **components**:

1. a ground extensional component

- a set of relational constructs for schema and data
- effectively modelled as graph data (or generalisations thereof)

2. an intensional component

- a set of inference rules (or other representations like GNNs or KGEs) over the constructs of the ground extensional component

3. a derived extensional component

- that can be produced applying inference rules over ground extensional components
- within the so-called *reasoning* process



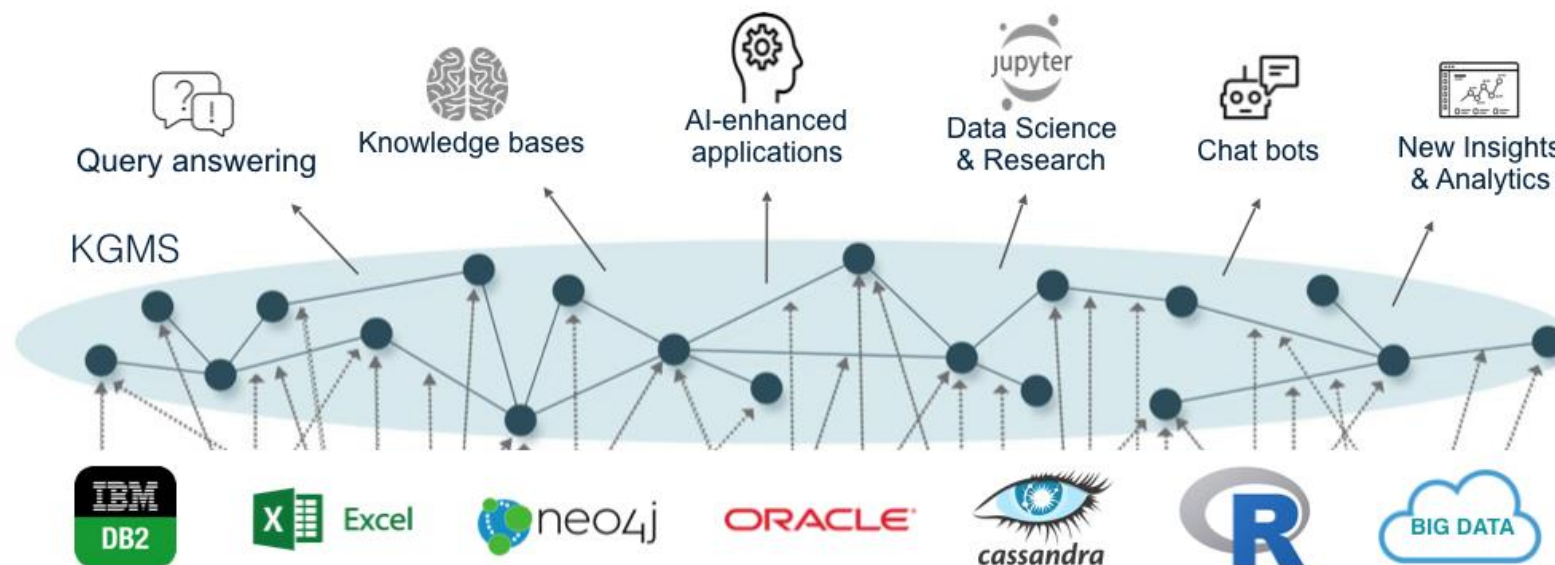
The Different Nature of KGs

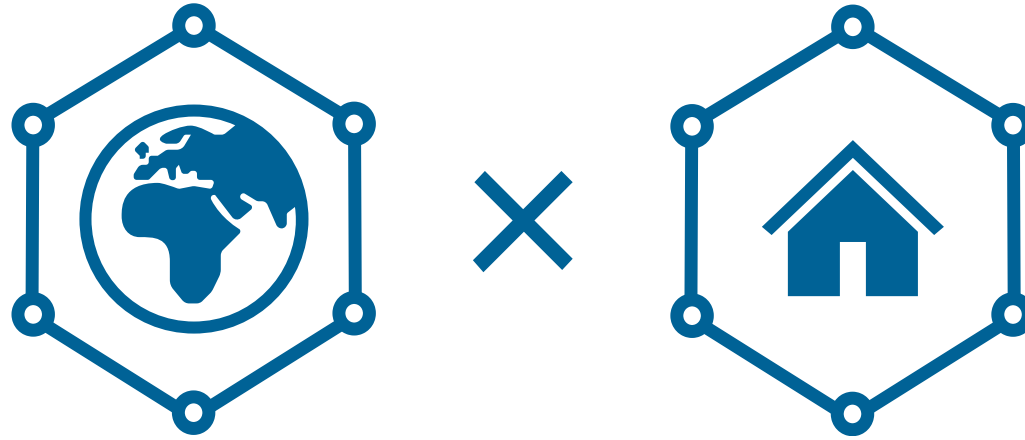
- **relational representation** of the schema-level constructs
 - suitable for a graph-based representation (e.g., property graph)
- **semi-structured** data model
 - same representation for both schema and data
- **inference rules or other reasoning capabilities**
 - embody conceptualisation of the domain knowledge



KG-based IT Architecture

- Enterprise AI-driven applications are typically written in procedural languages.
- They exploit KGs, declaratively defined with the representation language of the KGMS.
- Application logic of AI-driven applications **emerges from the combination of**
 - functionalities directly implemented **in the application code**
 - behaviour induced in the application **by external reasoning**
 - the balance depends on the specific application





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